Please attach this cover sheet to your completed homework

Homework 2 (Ch. 8) **ECE 311 – Hang Gao**

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Problem	Course outcome	Grade
8.1	1.c.	/36
8.2	1.c.	/20
8.5	1.c.	/15
8.6	1.c.	/15
8.11	1.c.	/14
	TOTAL:	/100

ABET Assessed Outcomes

1.c. Use appropriate models to formulate solutions related to electric circuit problems.

Homework #2
Due date: Sunday, February 8th by 11:59 PM (Electronically – Canvas)

8.1 A Y-connected balanced three-phase source is feeding a balanced three-phase load. The voltage and current of the source coil are

$$v(t) = 340\sin(377t + 0.5236)V$$

$$i(t) = 100\sin(377t + 0.87266)A$$

The phase shift angles in the two equations are in radians. Calculate the following:

a. The rms phase voltage
b. The rms line-to-line voltage
c. The rms current in the source
d. The rms current in the transmission line
e. The frequency of the supply
f. The power factor at the source side, state leading or lagging
g. The three-phase real power delivered to the load
h. The three-phase reactive power delivered to the load
i. If the load is connected in delta configuration, calculate the load impedance

8.2 The current and voltage of a wye-connected load are:

$$\overline{V}_{ca} = 480\angle -60° V$$

$$\overline{I}_b = 20\angle 120° A$$

a. Compute $\overline{V}_{an}$
b. Compute $\overline{I}_a$
c. Compute the power factor angle
d. Compute the real power of the load

8.5 A balanced wye-connected load of (4+j3)Ω is connected across a three-phase source of 173 V (line-to-line).

a. Find the transmission line current
b. Find the power factor, the complex power, the real power, and the reactive power of the load.
c. With V_{ab} as the reference, sketch the phasor diagram that shows all voltages and currents.

8.6 Two three-phase wye-connected loads are in parallel across a three-phase supply. The first load draws a current of 20 A at 0.9 power factor leading, and the second load draws a current of 30 A at 0.8 power factor lagging. Calculate the following:

a. The transmission line current
b. The power factor of the load
c. The real power supplied by the source if the line-to-line voltage of the transmission line is 400 V

8.11 A three-phase wye-connected source is energizing a delta-connected load. The phase voltage of the transmission line is $\overline{V}_{bn} = 120\angle 0° V$, and the total load impedance is $\overline{Z} = 9\angle 30° \Omega$. Compute the line current $\overline{I}_a$ and the real power consumed by the delta load.

8.1 **Note**: the following assumes that, unless otherwise notated, voltages and current magnitudes are given in rms values. Additionally, when calculating voltage, current, and impedance in *phasor* form, $\overline{V}_{an}$ is used as the reference voltage; that is, it is assumed to have a phase angle $\theta = 0°$. Finally, I have assumed throughout this problem that the wye-connected source is powering a delta-connected load.

a) For a sinusoidal voltage in a three-phase system given in the form $V_{\max} \sin(\omega t \pm \theta)$, the rms phase voltage V_{ph} is given by

$$V_{ph} = \frac{V_{\max}}{\sqrt{2}} = \frac{340}{\sqrt{2}} = 240.42 \text{ V}.$$

b) The line-to-line rms voltage in a three-phase, balanced, wye-connected source may be found using the phase voltages, where

$$\begin{aligned} \overline{V}_{an} &= V_{ph}\angle 0° \\ \overline{V}_{bn} &= V_{ph}\angle{-120°} \\ \overline{V}_{cn} &= V_{ph}\angle 120°, \end{aligned}$$

and

$$\begin{aligned} \overline{V}_{ab} &= \sqrt{3}V_{ph}\angle 30° \\ \overline{V}_{bc} &= \sqrt{3}V_{ph}\angle{-90°} \\ \overline{V}_{ca} &= \sqrt{3}V_{ph}\angle 150°. \end{aligned}$$

Therefore, solving just for the rms value of the line-to-line voltages,

$$V_{ab} = V_{bc} = V_{ca} = \sqrt{3}V_{ph} = 416.413 \text{ V}.$$

c) The rms source current through a three-phase, wye-connected, balanced source, I_a may be found as

$$\frac{I_{\max}}{\sqrt{2}} = \frac{100}{\sqrt{6}} = 70.711 \text{ A}$$

d) The rms value of the line current for any phase connected to a three-phase balanced load are found using a similar formula to phase voltage,

$$I_a = I_b = I_c = \frac{I_{max}}{\sqrt{2}} = \frac{100}{\sqrt{2}} = 70.711 \text{ A}.$$

e) The source's angular frequency in a three-phase balanced system is the same in all three lines, given in the time-dependent formula for voltage and current as ω, in radians per second. Converting $\omega = 377\frac{\text{rad}}{\text{s}}$ to units of frequency f in hertz,

$$f = \frac{\omega}{2\pi} = \frac{377}{2\pi} = 60 \text{ Hz}.$$

f) Power factor may be found as the cosine of the phase angle difference between voltage and current, $\theta_{VI} = \theta_V - \theta_I$,

$$P.F. = \cos(\theta_{VI}) = \cos(-0.3491) = 0.9397 \text{ (leading)},$$

since the phase angle difference θ_{VI} is negative, implying a capacitive load and a leading current waveform.

g) Three-phase real power $P = 3P_{ph} = 3V_{ph}I_{ph}\cos(\theta_{VI})$, so

$$P = (3 \times 240.416 \times 70.711)\cos(-0.3491) = 47.924 \text{ kW}.$$

h) Three-phase reactive power $Q = 3P_{ph}\cos(\theta_{IV})$, so

$$Q = (3 \times 240.416 \times 70.711)\sin(-0.3491) = -17.445 \text{ kVAr}.$$

i) Given a delta-connected three-phase balanced load, the load impedance $\overline{Z}$ may be found using

$$\overline{Z} = \frac{\overline{V}_{ab}}{\overline{I}_{ab}} = \frac{\overline{V}_{bc}}{\overline{I}_{bc}} = \frac{\overline{V}_{ca}}{\overline{I}_{ca}}.$$

Finding $\overline{I}_{ab}$, we may use the relationship

$$\overline{I}_a = \sqrt{3}\ \overline{I}_{ab}\angle{-30^\circ} \Rightarrow \overline{I}_{ab} = \frac{70.11}{\sqrt{3}}\angle 50^\circ = 40.83\angle 50^\circ \text{ A.}$$

Then,

$$\overline{Z} = \frac{416.41\angle 30^\circ}{40.83\angle 50^\circ} = 10.20\angle{-20^\circ}\ \Omega.$$

8.2 a) $\overline{V}_{an}$ may be found as

$$\overline{V}_{an} = V_{ph}\angle 0^\circ.$$

Thus, finding $\overline{V}_{an}$,

$$\sqrt{3}V_{ph}\angle 150^\circ = \overline{V}_{ca} \Rightarrow \overline{V}_{an} = \frac{480\angle{-60^\circ}}{\sqrt{3}\angle 150^\circ} = 277.13\angle 150^\circ \text{ V.}$$

b) $\overline{I}_a$ may be found as

$$\overline{I}_a = I_b\angle(\theta_{I_b} + 120^\circ) = 20\angle{-120^\circ} \text{ A.}$$

c) The phase angle difference $\theta_{VI} = -90^\circ$, so

$$P.F. = \cos(-1.5708) = 0 \text{ (leading)},$$

implying a purely capacitive load.

d) Real power P may be found as

$$P = 3P_{ph}\cos(\theta_{VI}) = (3 \times 277.128 \times 20)\cos(-90^\circ) = 0 \text{ W.}$$

8.5 a) Using $\overline{V}_{ab}$ as the reference voltage, with a phase angle of $\theta = 0^\circ$, we may find the voltage across the load as

$$\overline{V}_{an} = \frac{V_{ab}}{\sqrt{3}}\angle{-30^\circ} = 99.88\angle{-30^\circ} \text{ V,}$$

and the transmission line current as

$$\overline{I}_a = \frac{\overline{V}_{an}}{Z} = \frac{99.88\angle{-30^\circ}}{5\angle 36.87^\circ} = 19.98\angle{-66.87^\circ}\ \Omega$$

b) Given a phase angle difference $\theta_{VI} = 66.87^\circ$,

$$\begin{aligned}
&P.F. = \cos(1.1671) = 0.3928 \text{ (lagging)},\\
&S = P + jQ = 3V_{an}I_a = 3 \times 99.88 \times 19.98 = 5.987 \text{ kVA},\\
&P = S\cos(\theta_{VI}) = [5.987 \times 10^3]\cos(1.202) = 2.352 \text{ kW},\\
&Q = S\sin(\theta_{VI}) = [5.987 \times 10^3]\sin(1.202) = 5.506 \text{ kVAr}.
\end{aligned}$$

c) The following diagram was generated in MATLAB:

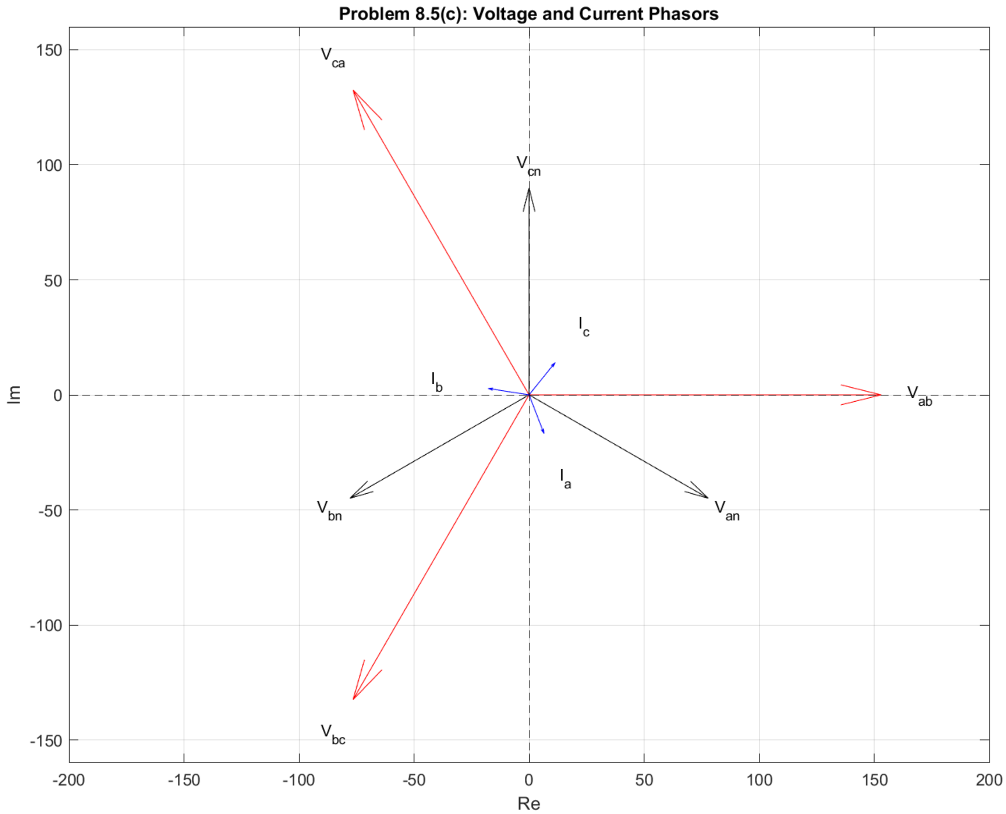

Figure 1: Voltage and Current Phasors

8.6 a) The total current through one phase A of the transmission line can be derived using KCL, and simply adding the current through the parallel loads together,

$$\overline{I}_a = 20\angle 25.84^\circ + 30\angle{-36.87^\circ} = 43.01\angle{-12.46^\circ} \text{ A}.$$

Then, the other two phases can be derived by simply shifting their phase angles 120° backwards,

$$\overline{I}_a = 43.01\angle{-12.46^\circ} \text{ A}$$
$$\overline{I}_b = 43.01\angle{-132.46^\circ} \text{ A}$$
$$\overline{I}_c = 43.01\angle 107.54^\circ \text{ A}.$$

b) Assuming a reference voltage phase angle of $\theta = 0^\circ$ for V_{an}, we may derive the power factor as

$$P.F. = \cos(\theta_{VI}) = \cos(0.2175) = 0.9764 \text{ (lagging)}.$$

c) Using the phase voltage $V_{ph} = \frac{400}{\sqrt{3}} = 230.94$ V, we may find real power as

$$P = [3 \times 230.94 \times 43.01] \cos(0.2175) = 29.096 \text{ kW}.$$

8.11 Given $\overline{V}_{bn} = 120\angle 0^\circ$ V, $\overline{V}_{an} = 120\angle 120^\circ$ V, and $\overline{V}_{ab} = 207.85\angle 150^\circ$ V. Then,

$$\overline{I}_{ab} = \frac{\overline{V}_{ab}}{\overline{Z}} = \frac{207.85\angle 150^\circ}{9\angle 30^\circ} = 23.09\angle 120^\circ \text{ A}.$$

Converting load current to transmission line current,

$$\overline{I}_a = \sqrt{3}\,\overline{I}_{ab}\angle{-30^\circ} = \sqrt{3} \times 23.09\angle 90^\circ = 39.99\angle 90^\circ \text{ A}.$$

Using the load current and voltage along with the phase angle difference to calculate real power consumed by the load,

$$3 I_{ab} V_{ab} \cos(0.5236) = 12.469 \text{ kW}.$$